

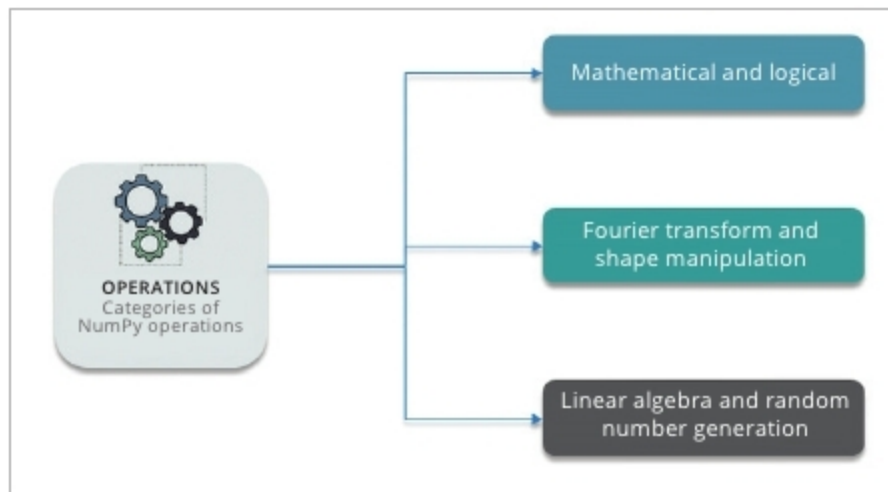


Figure 3: Operations performed by Python NumPy library  
(Image source: [googleimages.com](https://www.google.com/search?q=googleimages.com))

called features) that are intended to be non-redundant and informative, hence facilitating the subsequent learning as well as generalisation of steps (and in some conditions leading to better human interpretations). Feature extraction is basically related to dimensionality reduction. Python has a couple of image or video libraries that can assist in feature extraction.

### Different open source scientific Python libraries used for machine learning

There are many open source libraries used to implement ML for different applications. They are widely referred to as scientific Python libraries as they are put to use while performing the elementary machine learning tasks.

1. **Numpy:** This is a Python library that is widely used for N-dimensional array objects.
2. **Pandas:** This library file is used for Python data analysis, including different structures such as data frames.
3. **Matplotlib:** This is a 2D plotting library that produces publication-quality figures.
4. **Scikit-learn:** This library contains a couple of the ML learning algorithms that are used for data mining and data analysis tasks.
5. **Seaborn:** Sometimes it is difficult to get accurate plots with the help of Matplotlib as it focuses on line plots. In such cases, one can go with a more specific library, called Seaborn. It focuses on the visual aspects of different statistical models including heat maps, and also depicts the overall distribution of the data. **END** 🐧

### References

- [1] <http://www.wikipedia.org/>
- [2] <https://machinelearningmastery.com>
- [3] <https://www.forbes.com/sites/tiriasresearch>
- [4] <https://pythonprogramming.net>

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